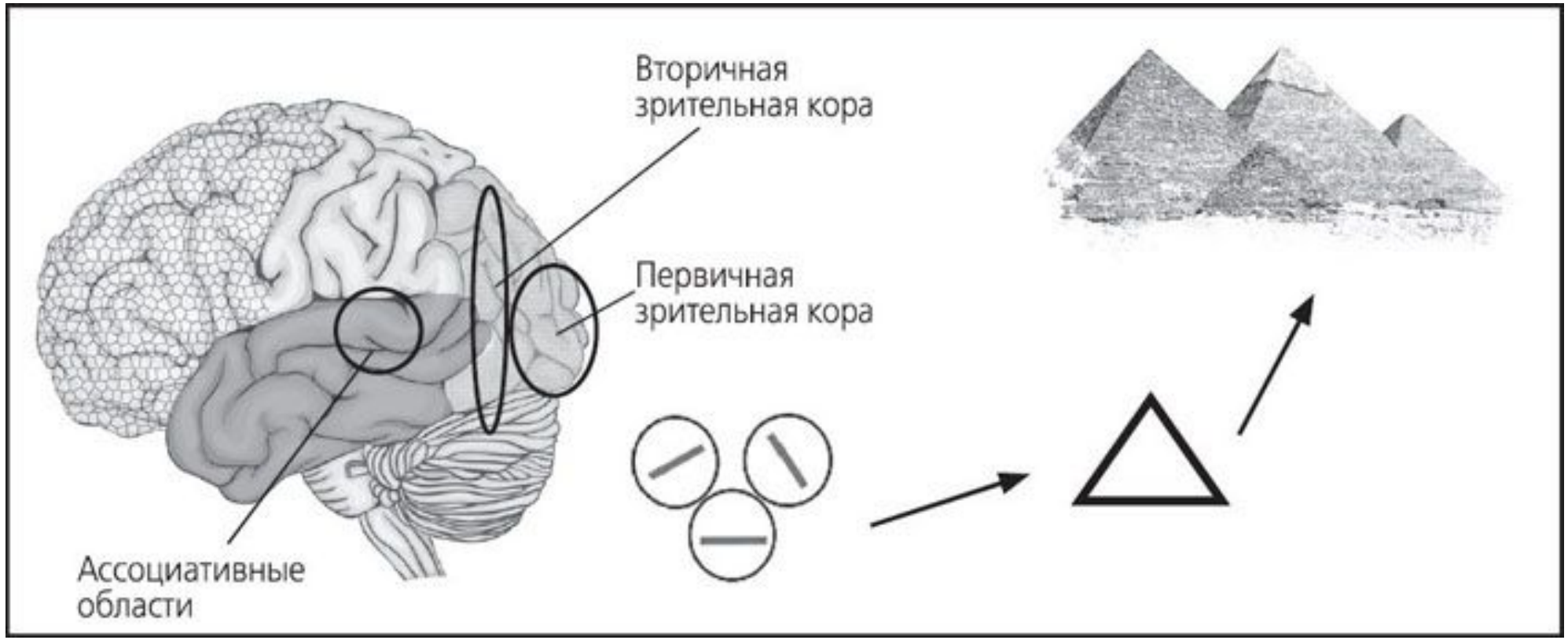
КУРС «НЕЙРОННЫЕ СЕТИ»

# Тема 4. Сверточные нейронные сети



Лекция 4 · Модуль 1

# Зрительная кора



# Ядра свертки

**X**

| | | | | | | |
|---|---|---|---|---|---|---|
| 11 | 12 | 13 | 14 | 15 | 16 | 17 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 |

**W**

| | | |
|---|---|---|
| 11 | 12 | 13 |
| 21 | 22 | 23 |
| 31 | 32 | 33 |

# Ядра свертки

**X**

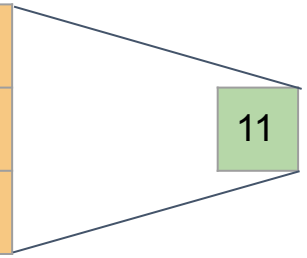
| | | | | | | |
|---|---|---|---|---|---|---|
| 11 | 12 | 13 | 14 | 15 | 16 | 17 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 |

$$z_{11}=w_{11}+x_{11}+w_{12}+x_{12}+\ldots+w_{33}+x_{33}+w_0$$

$$u_{11}=\sigma(z_{11})$$

**W**

| | | |
|---|---|---|
| 11 | 12 | 13 |
| 21 | 22 | 23 |
| 31 | 32 | 33 |



# Ядра свертки

**X**

| | | | | | | |
|---|---|---|---|---|---|---|
| 11 | 12 | 13 | 14 | 15 | 16 | 17 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 |

$$z_{12}=w_{11}+x_{12}+w_{12}+x_{13}+\ldots+w_{33}+x_{34}+w_0$$

$$u_{12}=\sigma(z_{12})$$

**W**

| | | |
|---|---|---|
| 11 | 12 | 13 |
| 21 | 22 | 23 |
| 31 | 32 | 33 |



# Ядра свертки

**X**

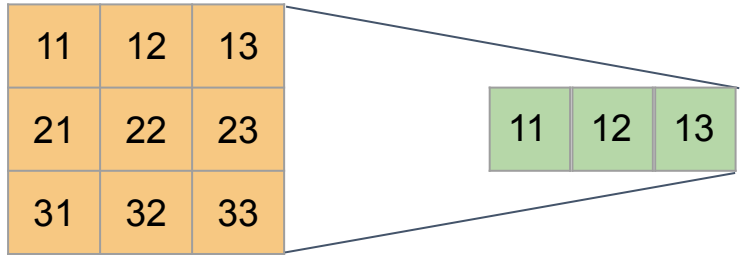
| | | | | | | |
|---|---|---|---|---|---|---|
| 11 | 12 | 13 | 14 | 15 | 16 | 17 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 |

$$z_{13}=w_{11}+x_{13}+w_{14}+x_{13}+\ldots+w_{33}+x_{35}+w_0$$

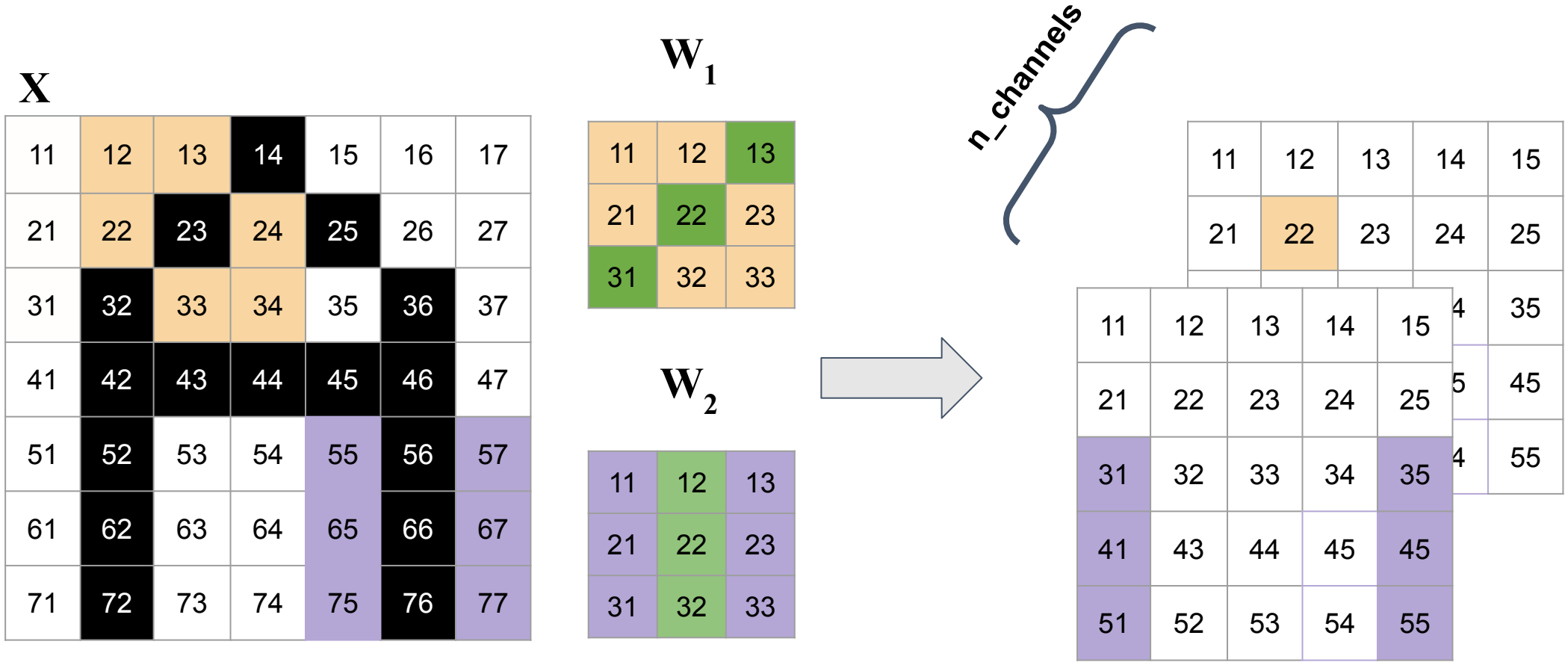
$$u_{13}=\sigma(z_{13})$$

**W**

| | | |
|---|---|---|
| 11 | 12 | 13 |
| 21 | 22 | 23 |
| 31 | 32 | 33 |



# Ядра свертки



# Ядра свертки

**width = 7**

**height = 1**

| | | | | | | |
|---|---|---|---|---|---|---|
| 11 | 12 | 13 | 14 | 15 | 16 | 17 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 |

По умолчанию ядра сжимают размерность исходного изображения

| | | | | |
|---|---|---|---|---|
| 11 | 12 | 13 | 14 | 15 |
| 21 | 22 | 23 | 24 | 25 |
| 31 | 32 | 33 | 34 | 35 |
| 41 | 43 | 44 | 45 | 45 |
| 51 | 52 | 53 | 54 | 55 |

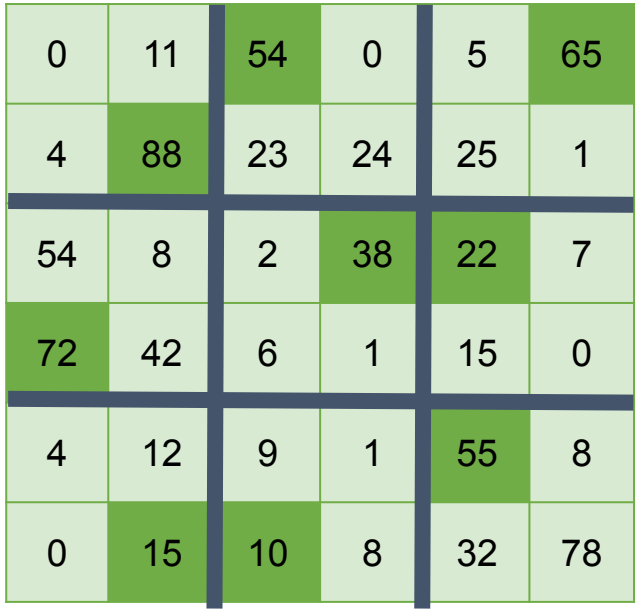
# Ядра свертки

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 0 |
| 0 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 0 |
| 0 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 0 |
| 0 | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 0 |
| 0 | 51 | 52 | 53 | 54 | 55 | 56 | 57 | 0 |
| 0 | 61 | 62 | 63 | 64 | 65 | 66 | 67 | 0 |
| 0 | 71 | 72 | 73 | 74 | 75 | 76 | 77 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

**Padding** - способ сохранить исходную размерность

| | | | | | | |
|---|---|---|---|---|---|---|
| 11 | 12 | 13 | 14 | 15 | 16 | 17 |
| 21 | 22 | 23 | 24 | 25 | 26 | 27 |
| 31 | 32 | 33 | 34 | 35 | 36 | 37 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 |
| 51 | 52 | 53 | 54 | 55 | 56 | 57 |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 |
| 71 | 72 | 73 | 74 | 75 | 76 | 77 |

# MaxPooling, MinPooling, AveragePooling




**MaxPooling** - отбор наибольших значений

**MinPooling** - отбор наименьших значений

**AveragePooling** - отбор средних значений

# MaxPooling, MinPooling, AveragePooling

| | | | | | |
|---|---|---|---|---|---|
| 0 | 11 | 54 | 0 | 5 | 65 |
| 4 | 88 | 23 | 24 | 25 | 1 |
| 54 | 8 | 2 | 38 | 22 | 7 |
| 72 | 42 | 6 | 1 | 15 | 0 |
| 4 | 12 | 9 | 1 | 55 | 8 |
| 0 | 15 | 10 | 8 | 32 | 78 |

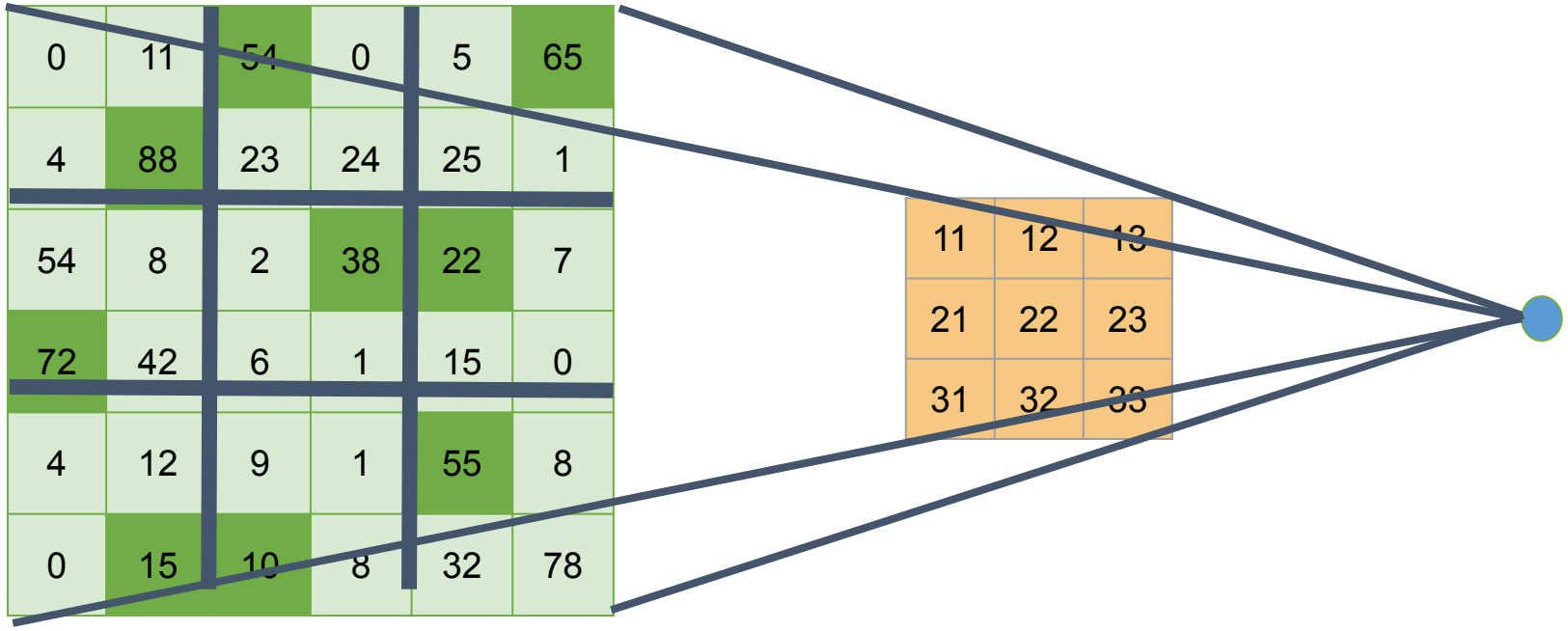
**MaxPooling** - отбор наибольших значений

**MinPooling** - отбор наименьших значений

**AveragePooling** - отбор средних значений



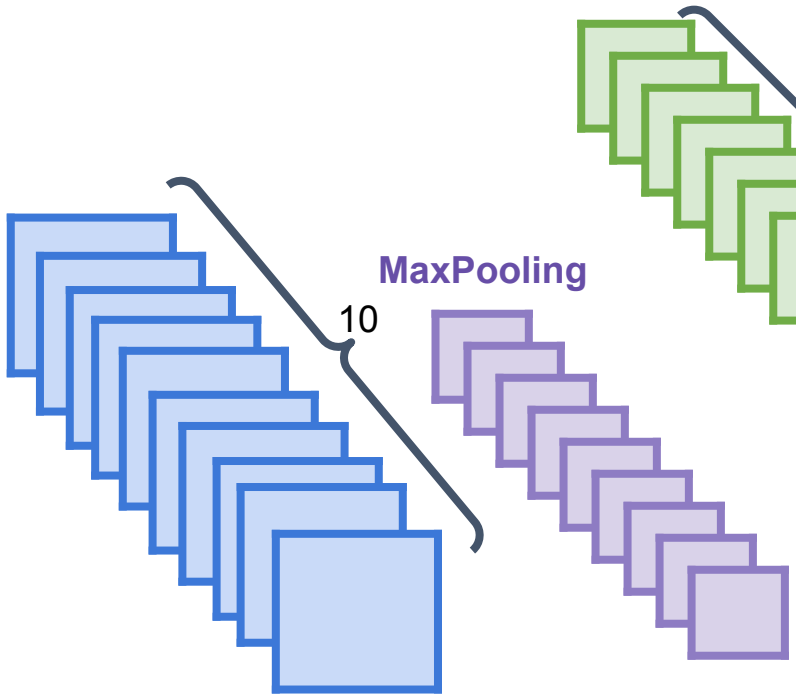
# MaxPooling, MinPooling, AveragePooling



# Общая архитектура



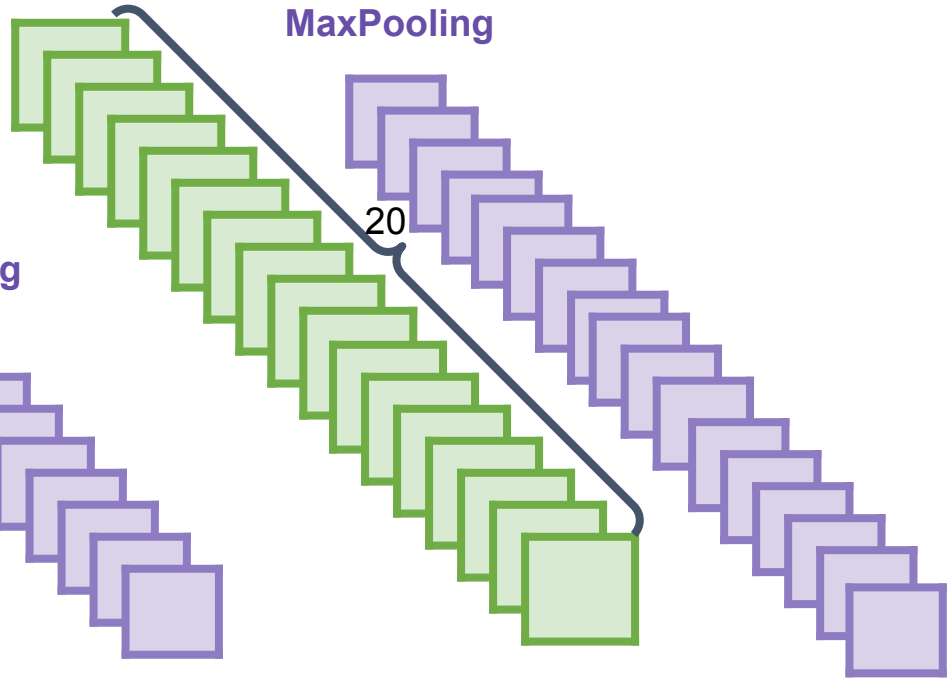

(10, 32, 32)




(10, 32, 32)

MaxPooling

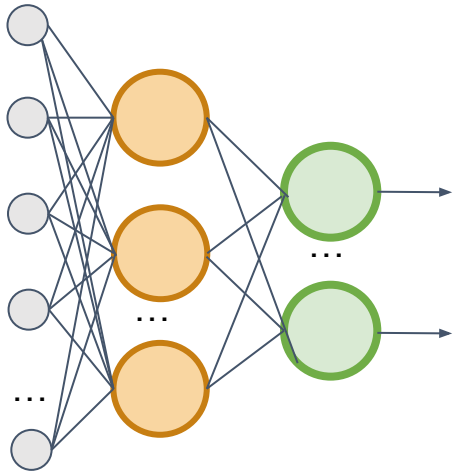
(10, 16, 16)




MaxPooling

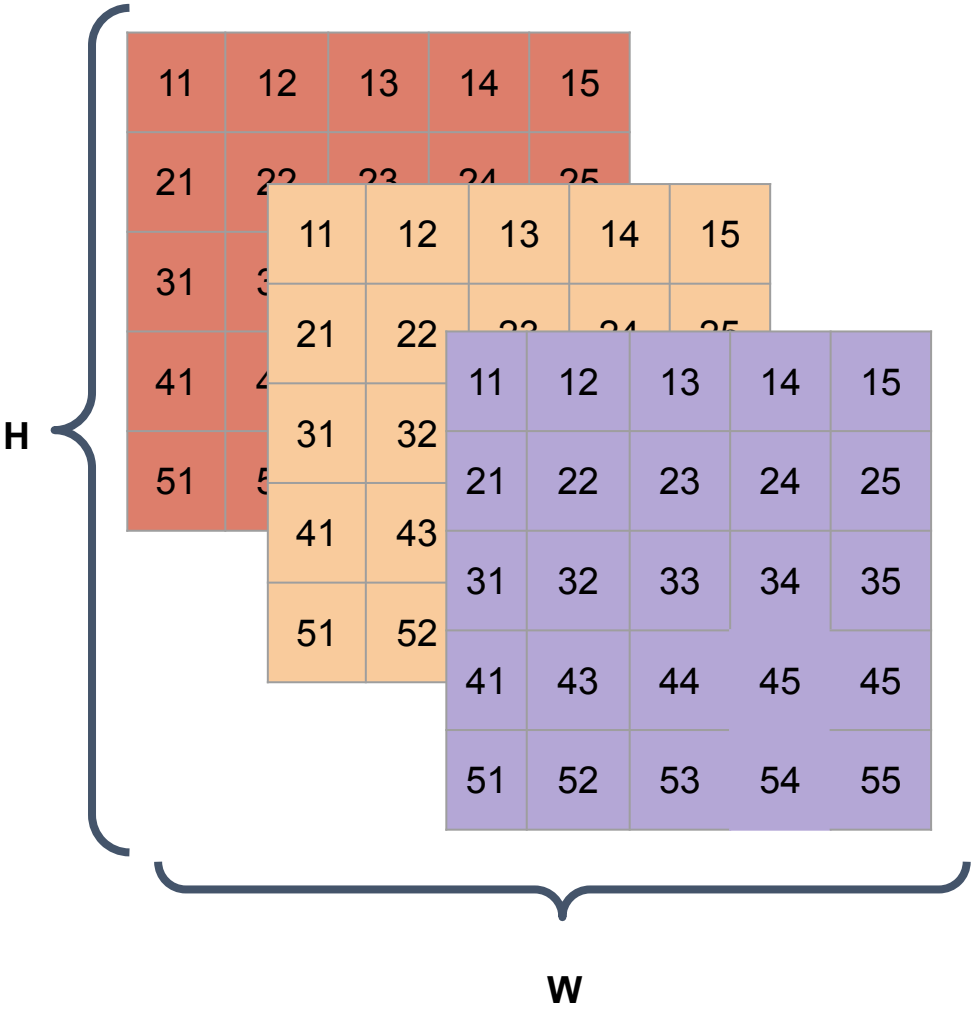
(20, 16, 16)

(20, 8, 8)



1280

# Класс nn.Conv2d




**nn.Conv1d** - для одномерной свертки

**nn.Conv2d** - для двумерной свертки (изображение)

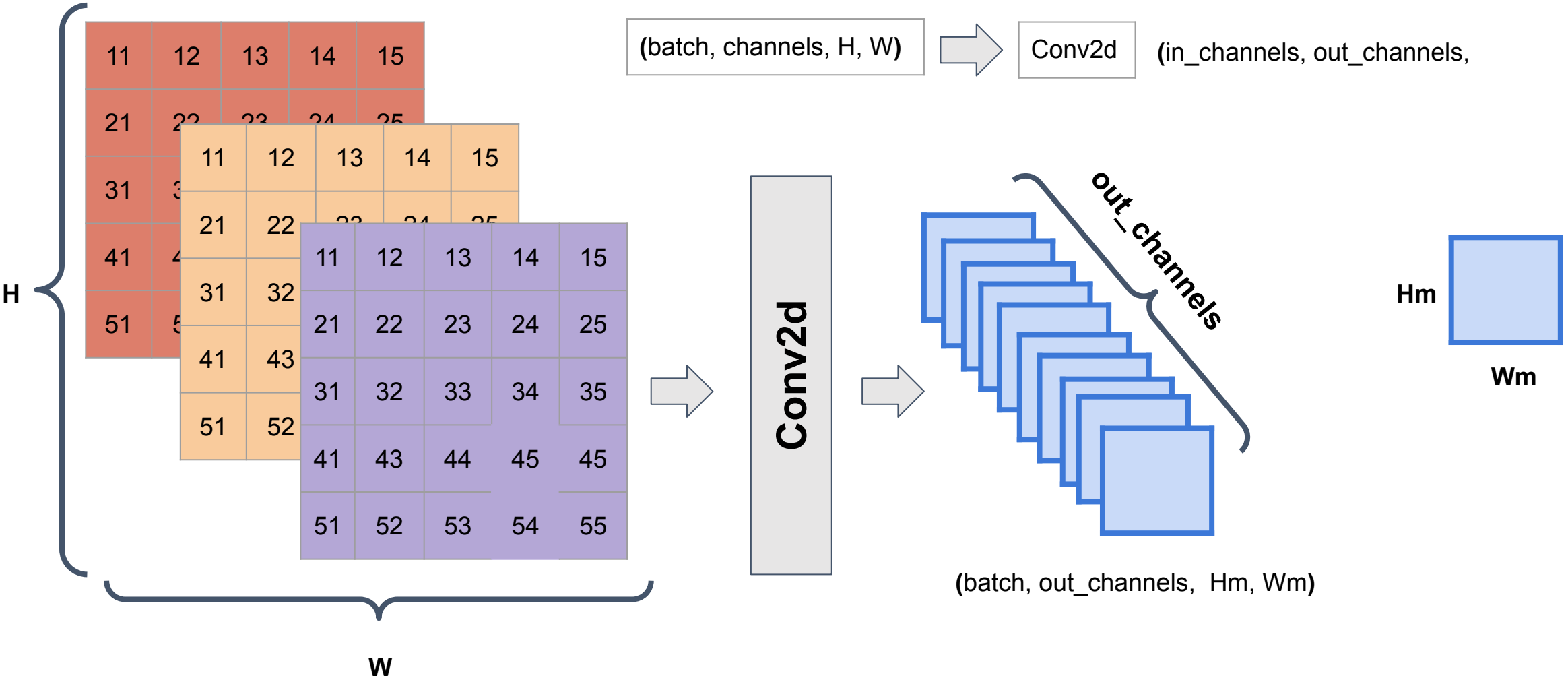
**nn.Conv3d** - для трехмерной свертки (снимок МРТ)

**nn.MaxPool1d** - для одномерного канала

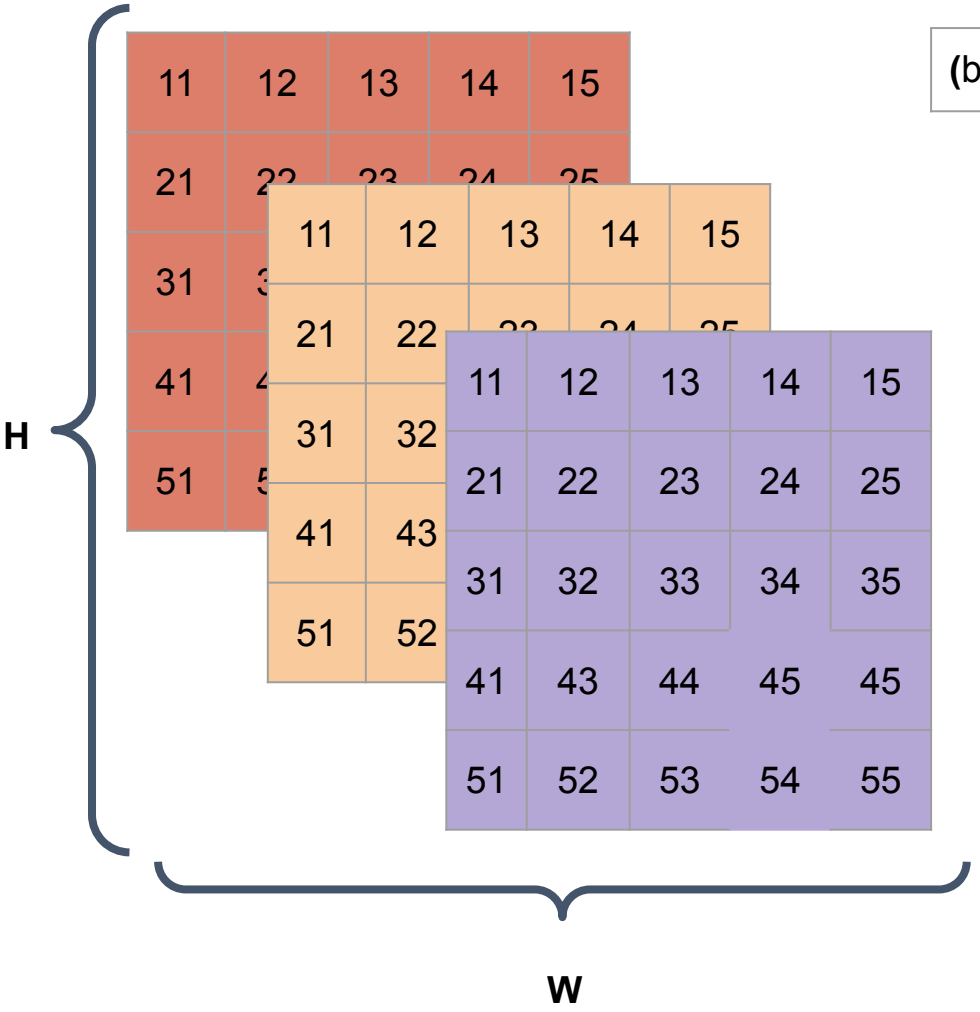
**nn.MaxPool2d** - для двумерного канала

**nn.MaxPool3d** - для трехмерного канала

# Класс nn.Conv2d



# Класс nn.Conv2d




(batch, channels, H, W) → Conv2d (in_channels, out_channels, kernel_size

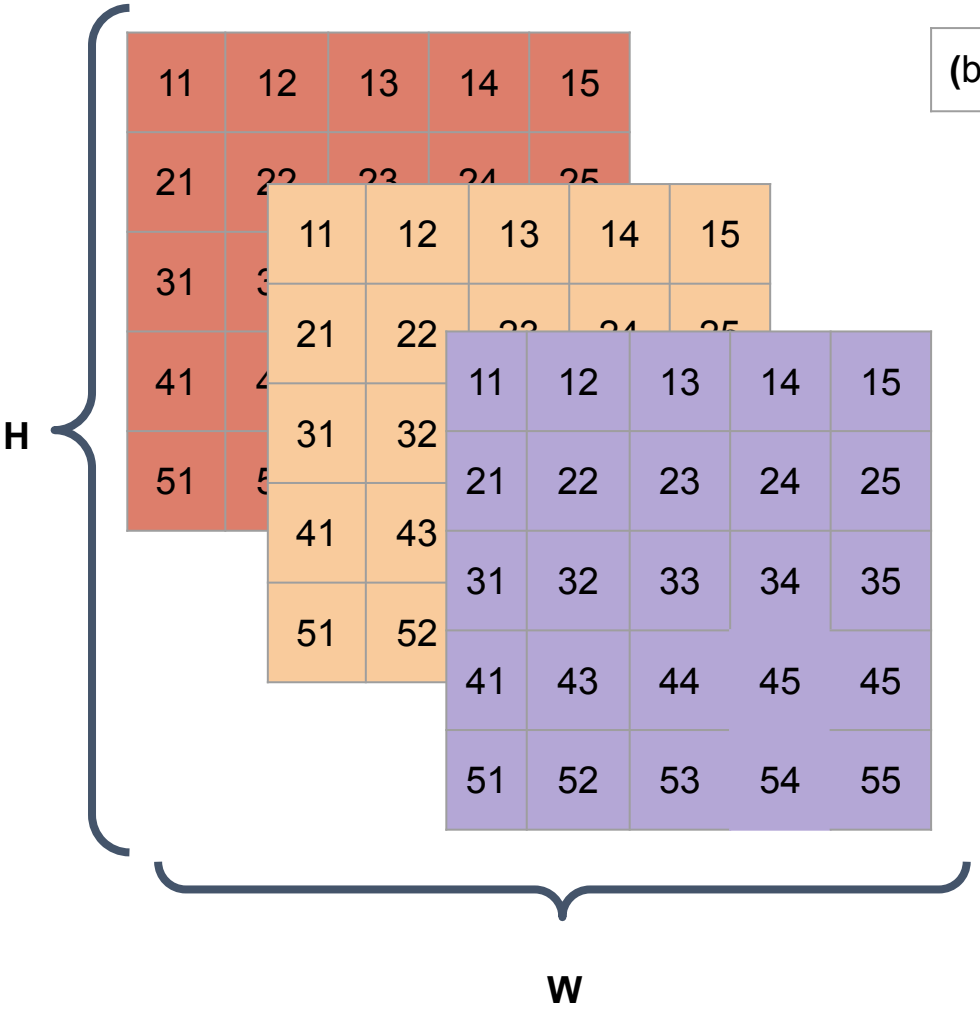
| | | |
|---|---|---|
| 11 | 12 | 13 |
| 21 | 22 | 23 |
| 31 | 32 | 33 |

**kernel_size = 3**
**kernel_size = (3, 3)**

| | | |
|---|---|---|
| 11 | 12 | 13 |
| 21 | 22 | 23 |

**kernel_size = (2, 3)**

# Класс nn.Conv2d




(batch, channels, H, W) → Conv2d (in_channels, out_channels, kernel_size, stride=1, padding=0)

| | | |
|---|---|---|
| 11 | 12 | 13 |
| 21 | 22 | 23 |
| 31 | 32 | 33 |

**kernel_size = 3**
**kernel_size = (3, 3)**

| | | |
|---|---|---|
| 11 | 12 | 13 |
| 21 | 22 | 23 |

**kernel_size = (2, 3)**

**padding**='valid' - то же самое что и padding=0
**padding**='same'

# Класс nn.Conv2d

(batch, channels, H, W) → MaxPool2d (kernel_size, stride=None, …)

| | | | | | |
|---|---|---|---|---|---|
| 0 | 11 | 54 | 0 | 5 | 65 |
| 4 | 88 | 23 | 24 | 25 | 1 |
| 54 | 8 | 2 | 38 | 22 | 7 |
| 72 | 42 | 6 | 1 | 15 | 0 |
| 4 | 12 | 9 | 1 | 55 | 8 |
| 0 | 15 | 10 | 8 | 32 | 78 |

torch.nn.MaxPool2d(kernel_size, stride=None, padding=0, dilation=1, return_indices=False, ceil_mode=False)

# Класс nn.Conv2d

(batch, channels, H, W) → MaxPool2d (kernel_size, stride=None, …)

| | | | | | |
|---|---|---|---|---|---|
| 0 | 11 | 54 | 0 | 5 | 65 |
| 4 | 88 | 23 | 24 | 25 | 1 |
| 54 | 8 | 2 | 38 | 22 | 7 |
| 72 | 42 | 6 | 1 | 15 | 0 |
| 4 | 12 | 9 | 1 | 55 | 8 |
| 0 | 15 | 10 | 8 | 32 | 78 |

→ **MaxPool2d** →

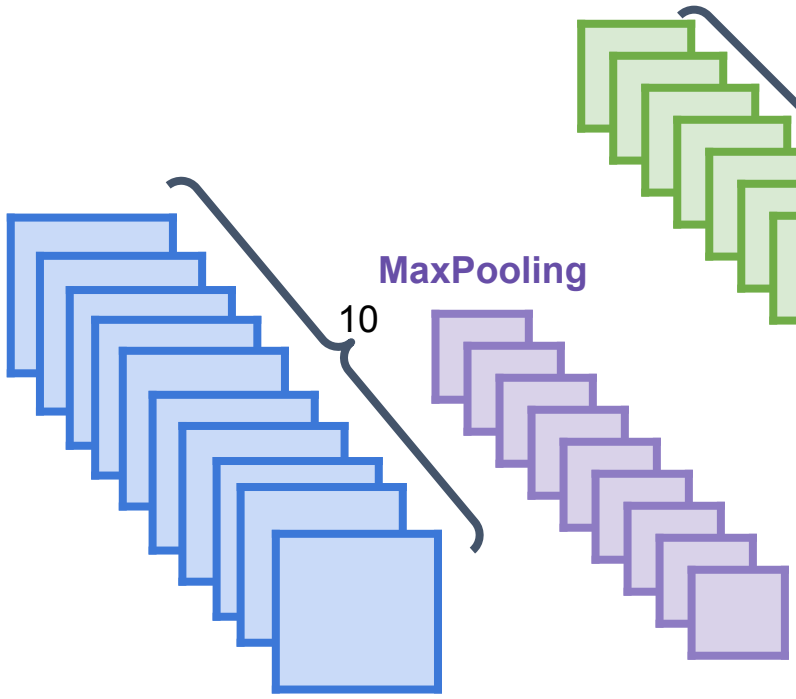
| | | |
|---|---|---|
| 88 | 54 | 65 |
| 72 | 38 | 22 |
| 15 | 10 | 55 |

(batch, channels, H // 2, W // 2)

# Общая архитектура



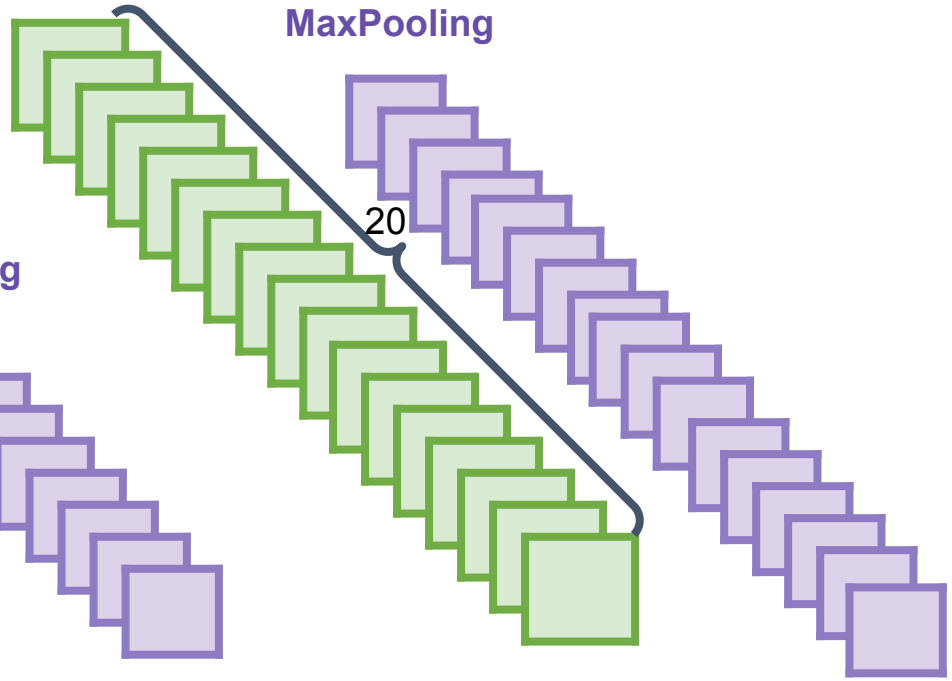

(10, 32, 32)




(10, 32, 32)

MaxPooling

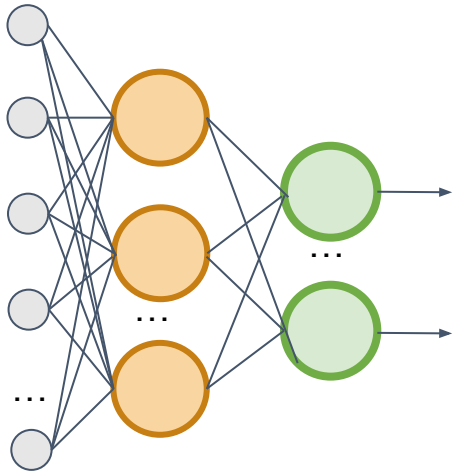
(10, 16, 16)




MaxPooling

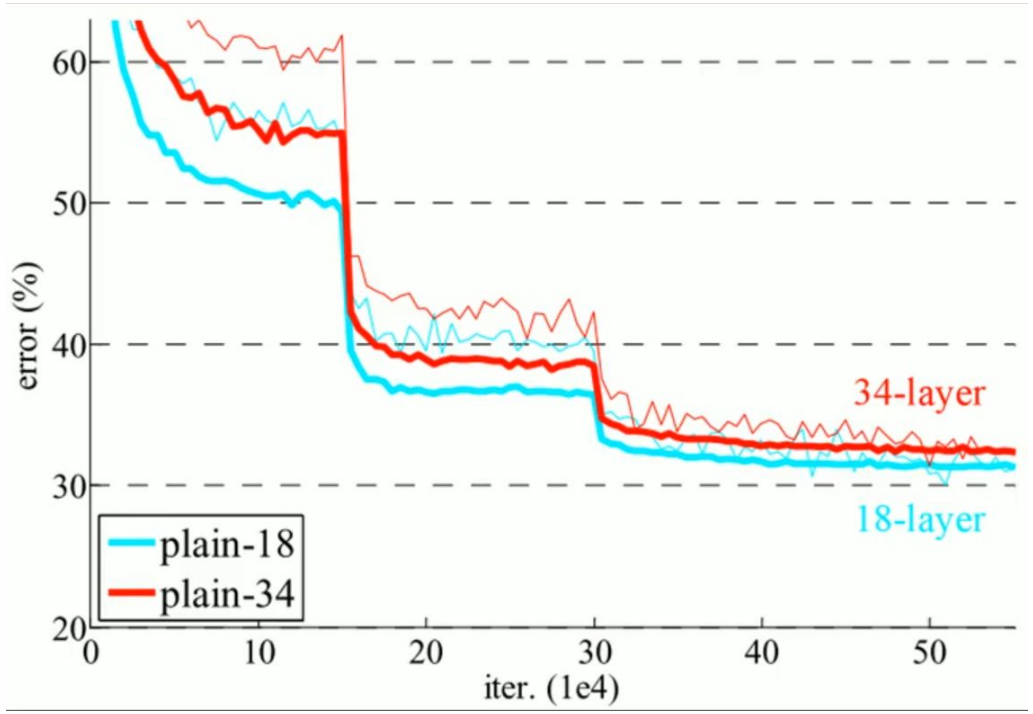
(20, 16, 16)

(20, 8, 8)



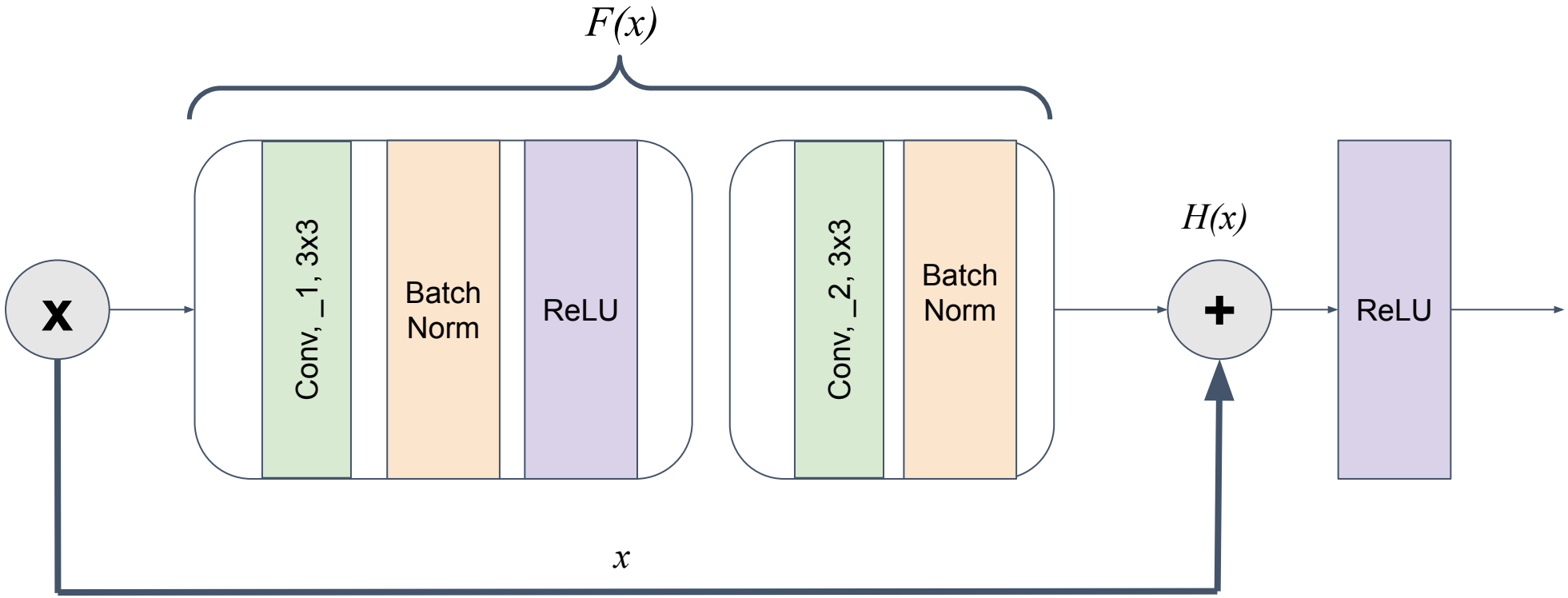
1280

# Проблема затухающих градиентов

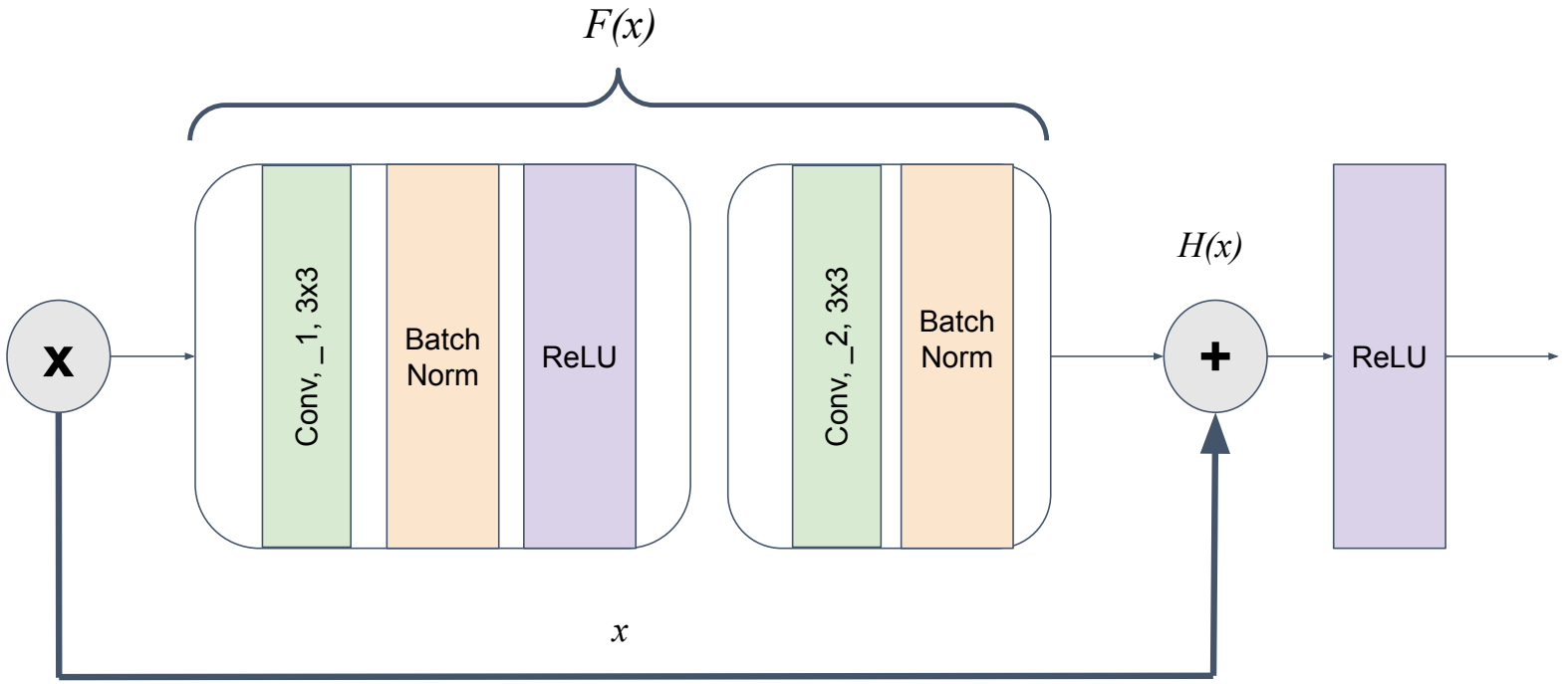



**Проблема затухающих (исчезающих) градиентов (vanishing gradients)**

# Архитектура ResNet



# Архитектура ResNet



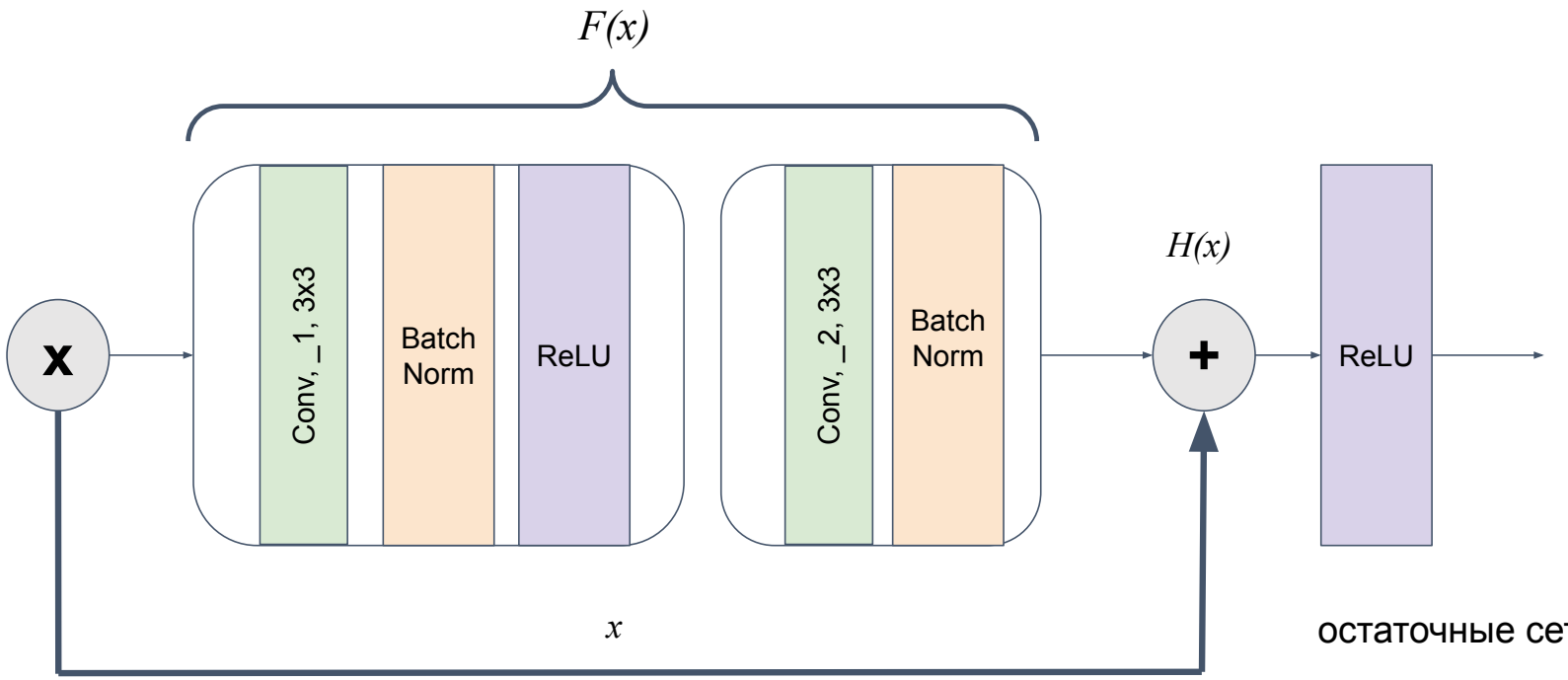

$$H(x) = F(x) + x$$

$$y = F(x,\ W)$$

$$F(x) = H(x) - x$$

$$\frac{\partial H(x)}{\partial x} = 1 + \frac{\partial F(x)}{\partial x}$$

# Базовый блок




$$H(x) = F(x) + x$$

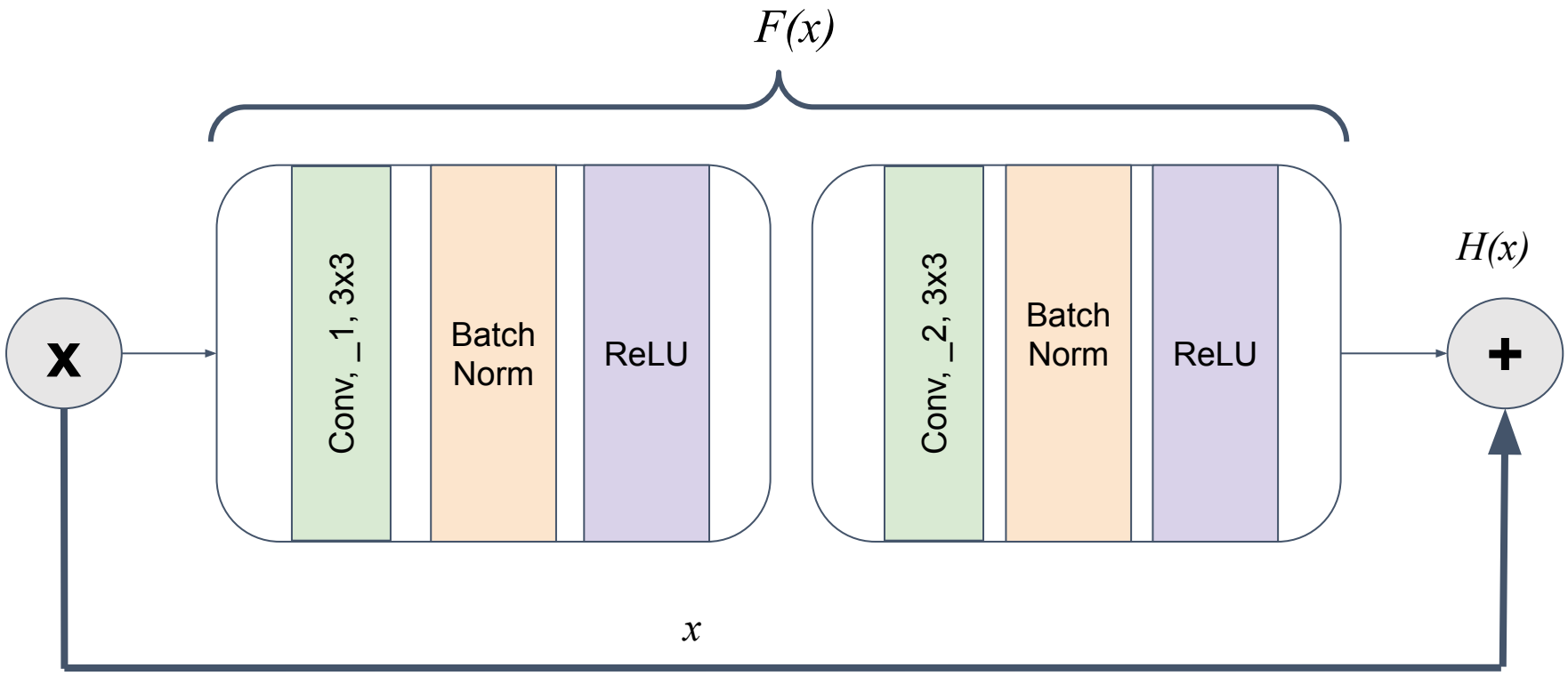
$$y = F(x, W)$$

$$F(x) = H(x) - x$$

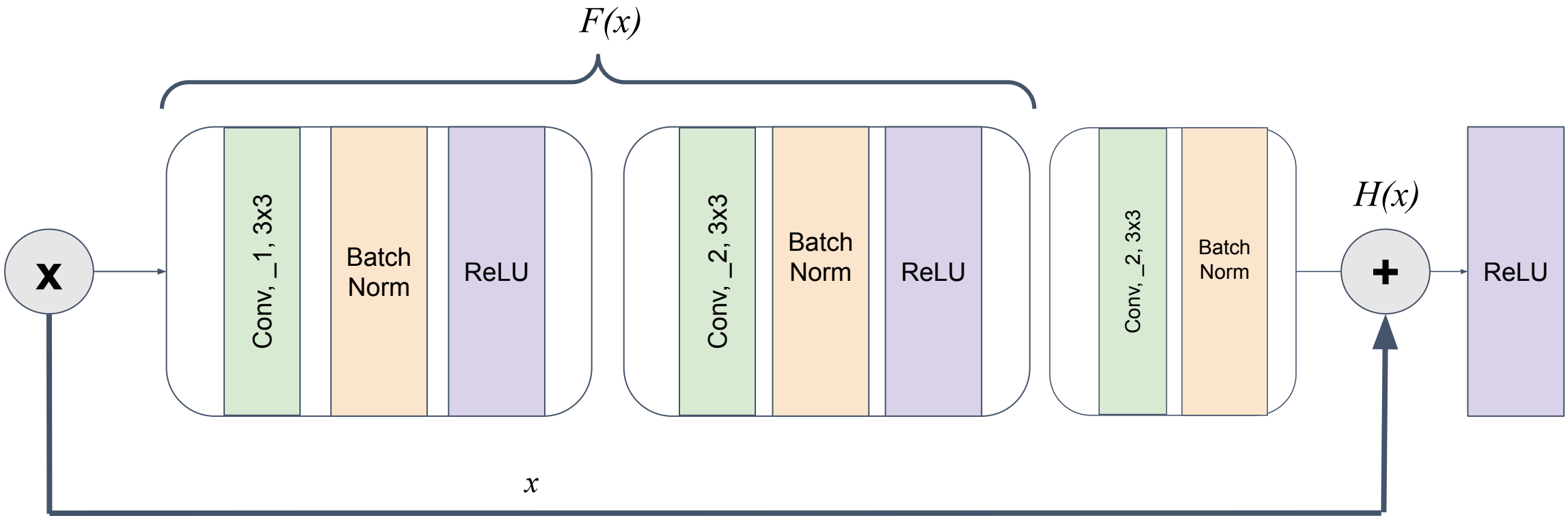
$$\frac{\partial H(x)}{\partial x} = 1 + \frac{\partial F(x)}{\partial x}$$

остаточные сети (residual networks) - ResNet (152 слоя)

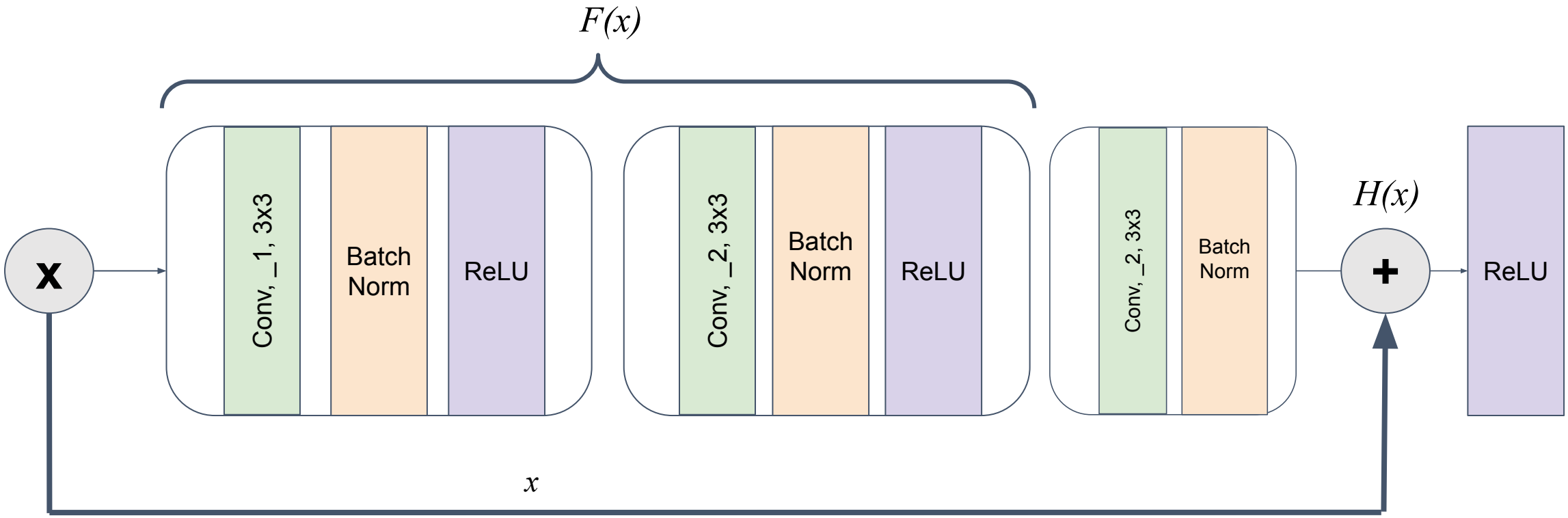
# Базовый блок v2



# Bottleneck Block



# Bottleneck Block




1-й слой: 1 * 1 * 128 * 64 = 8192
2-й слой: 1 * 1 * 64 * 64 = 36864
3-й слой: 1 * 1 * 64 * 128 = 8192

итого: 8192+36864+8192 = 53248

3 * 3 * 128 * 128 = 147456

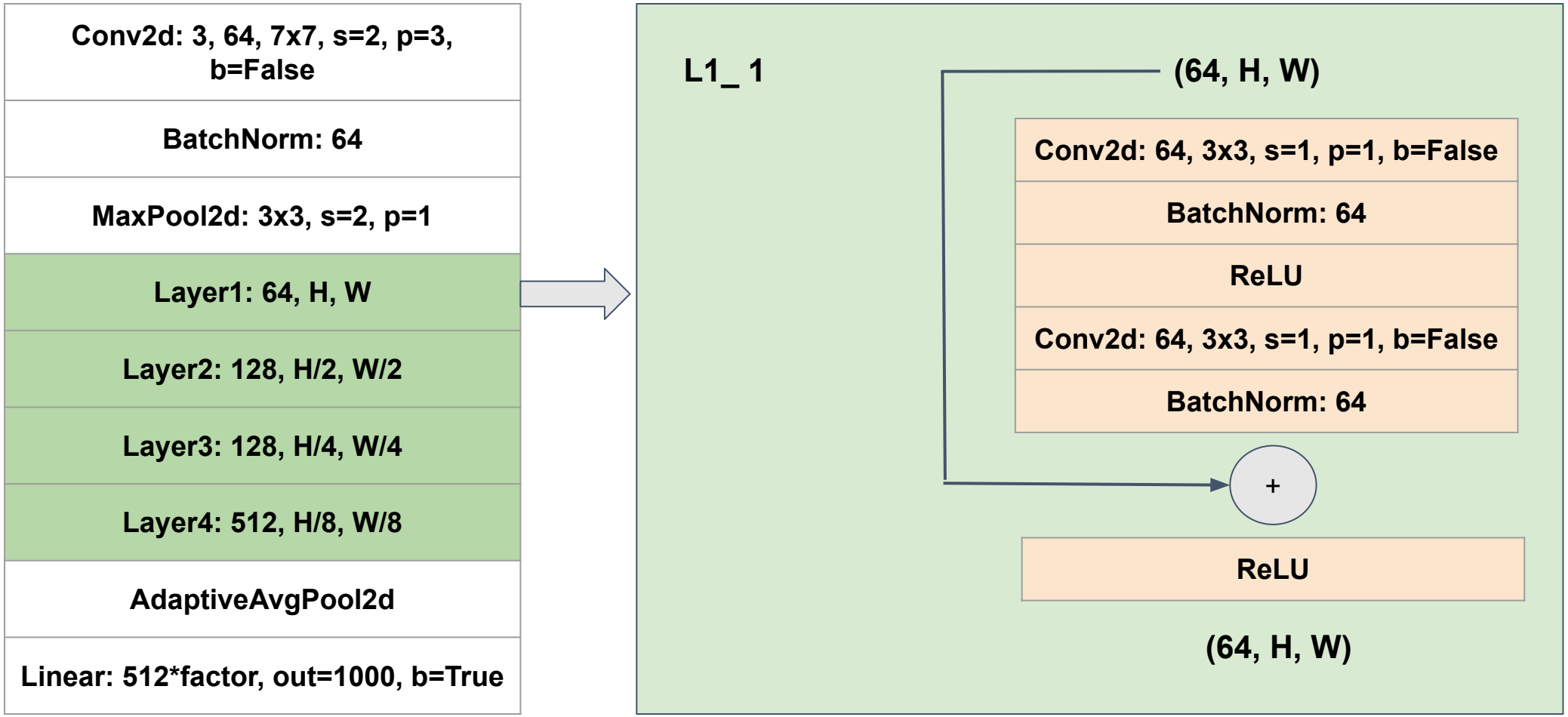
# Общая архитектура сетей ResNet

| Conv2d: 3, 64, 7x7, s=2, p=3, b=False |
|---|
| BatchNorm: 64 |
| MaxPool2d: 3x3, s=2, p=1 |
| Layer1: 64, H, W |
| Layer2: 128, H/2, W/2 |
| Layer3: 128, H/4, W/4 |
| Layer4: 512, H/8, W/8 |
| AdaptiveAvgPool2d |
| Linear: 512*factor, out=1000, b=True |

| BasicBlock | |
|---|---|
| resnet18 | resnet34 |
| 2 | 3 |
| 2 | 4 |
| 2 | 6 |
| 2 | 3 |

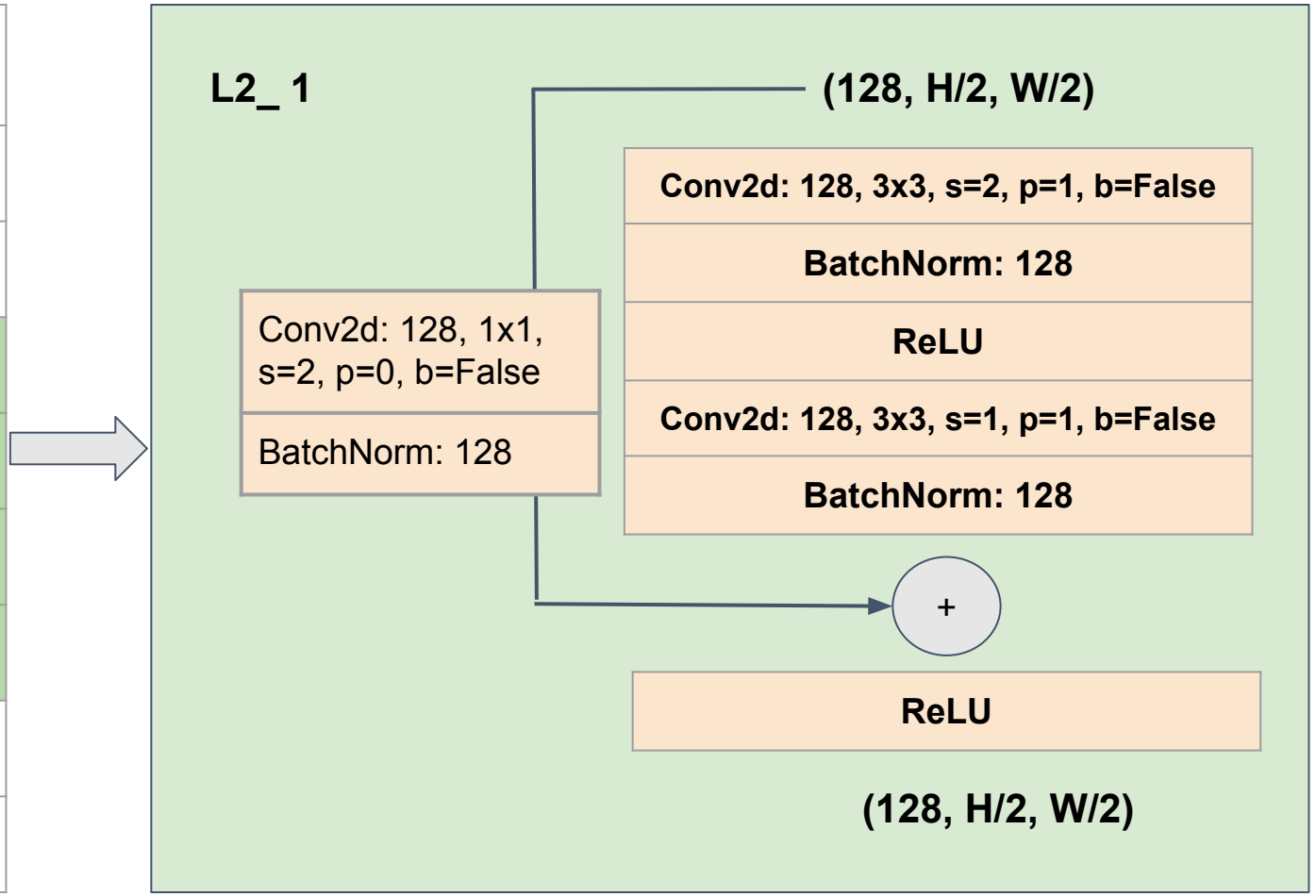
| BasicBlock | |
|---|---|
| resnet50 | resnet101 |
| 3 | 3 |
| 4 | 4 |
| 6 | 23 |
| 3 | 3 |

# Архитектура сети ResNet-18



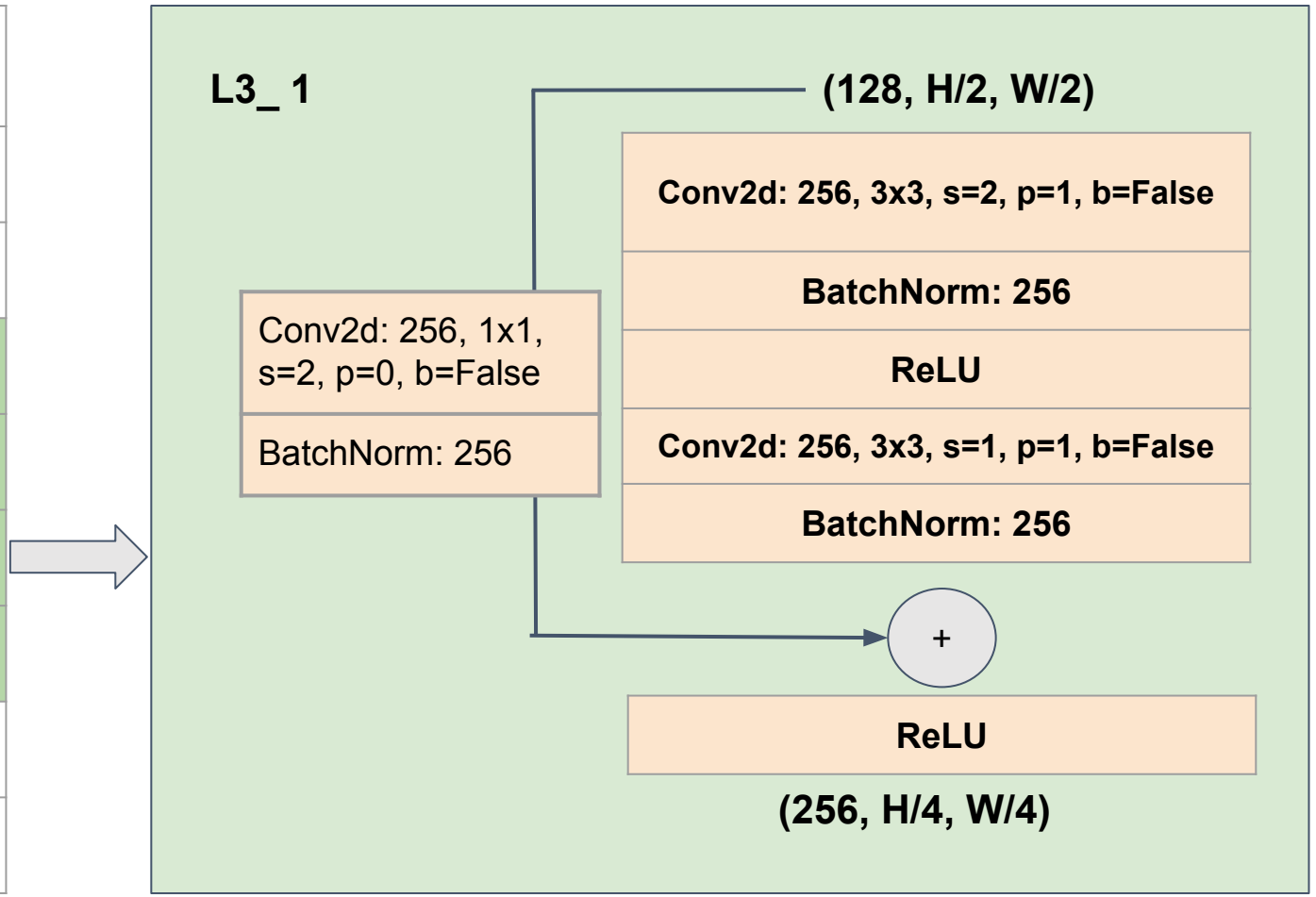
# Архитектура сети ResNet-18

| |
|---|
| Conv2d: 3, 64, 7x7, s=2, p=3, b=False |
| BatchNorm: 64 |
| MaxPool2d: 3x3, s=2, p=1 |
| Layer1: 64, H, W |
| Layer2: 128, H/2, W/2 |
| Layer3: 128, H/4, W/4 |
| Layer4: 512, H/8, W/8 |
| AdaptiveAvgPool2d |
| Linear: 512*factor, out=1000, b=True |

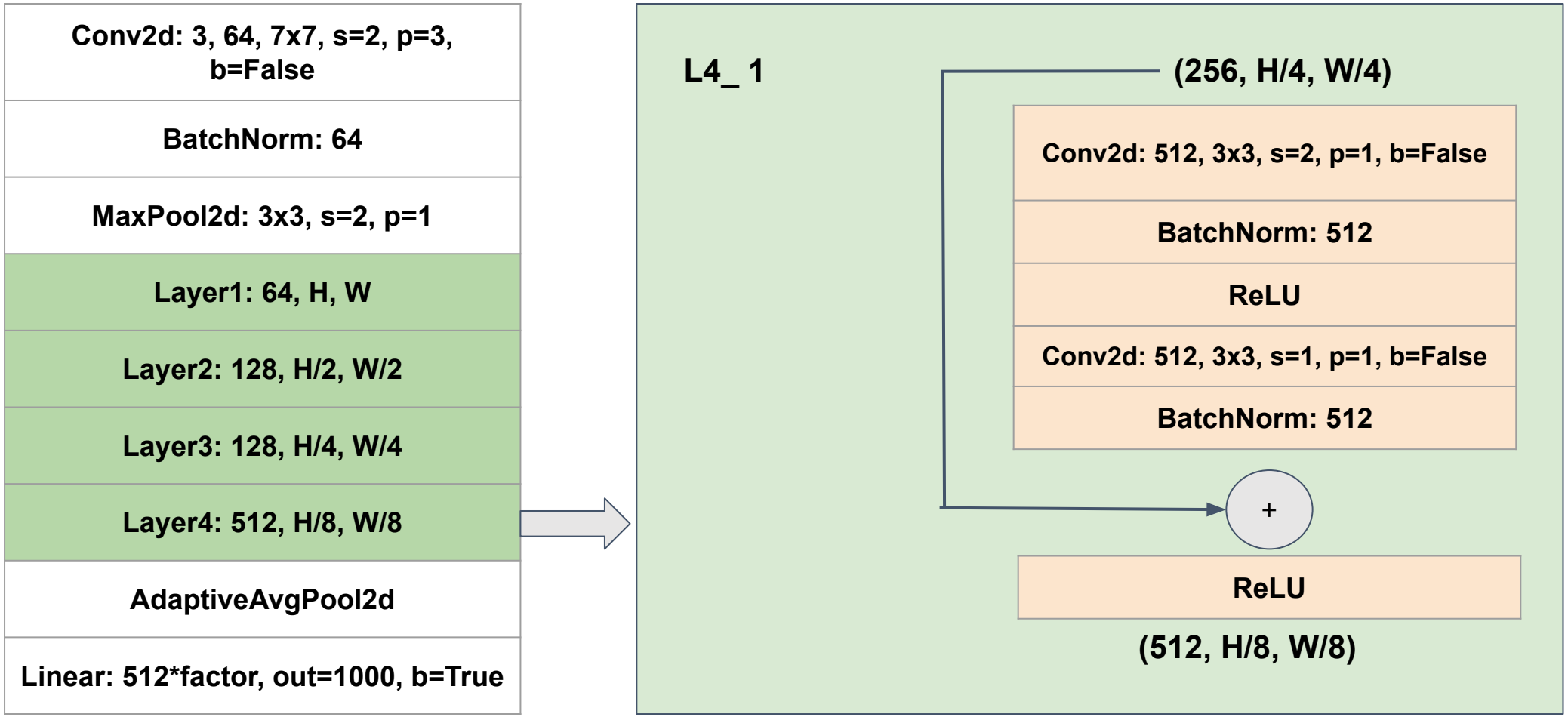


# Архитектура сети ResNet-18

| |
|---|
| Conv2d: 3, 64, 7x7, s=2, p=3, b=False |
| BatchNorm: 64 |
| MaxPool2d: 3x3, s=2, p=1 |
| Layer1: 64, H, W |
| Layer2: 128, H/2, W/2 |
| Layer3: 128, H/4, W/4 |
| Layer4: 512, H/8, W/8 |
| AdaptiveAvgPool2d |
| Linear: 512*factor, out=1000, b=True |

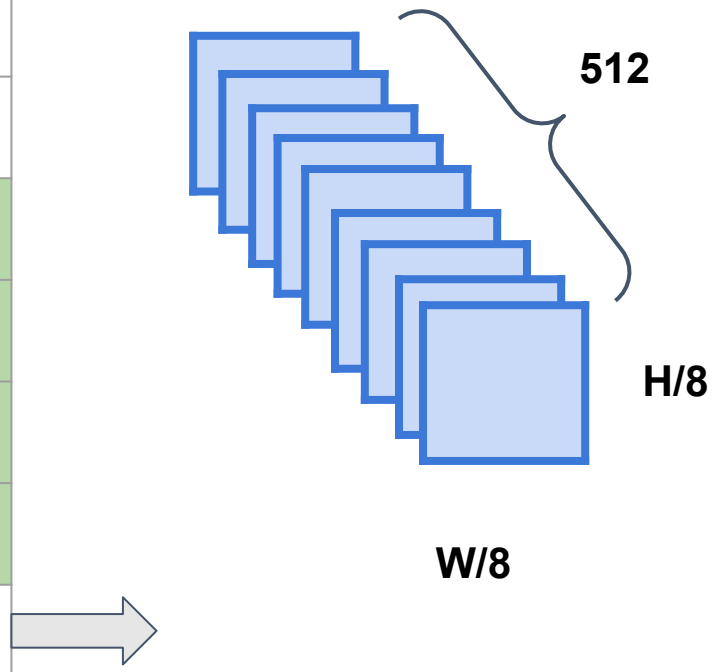


# Архитектура сети ResNet-18



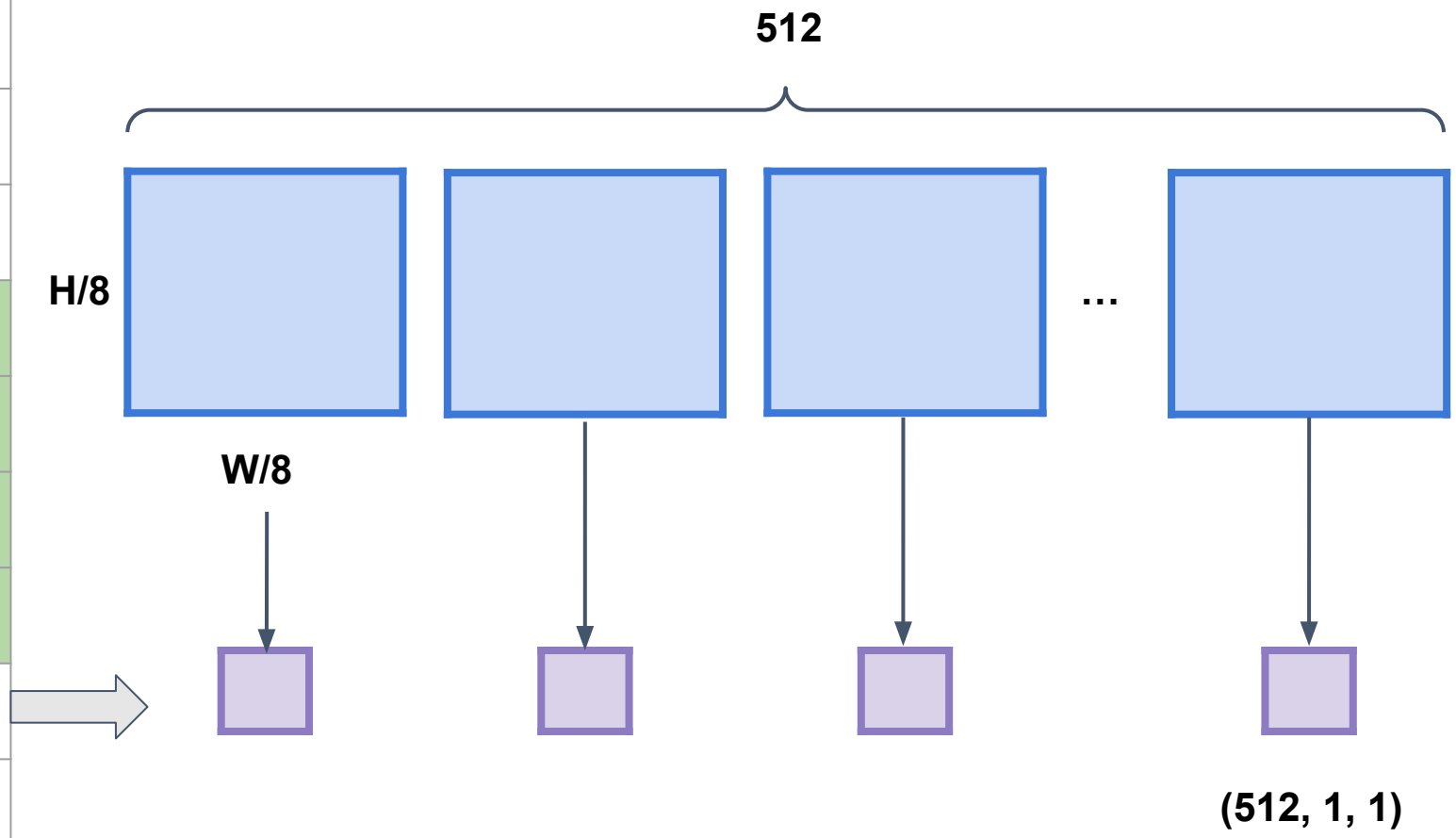
# Архитектура сети ResNet-18

| |
|---|
| Conv2d: 3, 64, 7x7, s=2, p=3, b=False |
| BatchNorm: 64 |
| MaxPool2d: 3x3, s=2, p=1 |
| Layer1: 64, H, W |
| Layer2: 128, H/2, W/2 |
| Layer3: 128, H/4, W/4 |
| Layer4: 512, H/8, W/8 |
| AdaptiveAvgPool2d |
| Linear: 512*factor, out=1000, b=True |



# Архитектура сети ResNet-18

| |
|---|
| Conv2d: 3, 64, 7x7, s=2, p=3, b=False |
| BatchNorm: 64 |
| MaxPool2d: 3x3, s=2, p=1 |
| Layer1: 64, H, W |
| Layer2: 128, H/2, W/2 |
| Layer3: 128, H/4, W/4 |
| Layer4: 512, H/8, W/8 |
| AdaptiveAvgPool2d |
| Linear: 512*factor, out=1000, b=True |



# Общая архитектура сетей ResNet

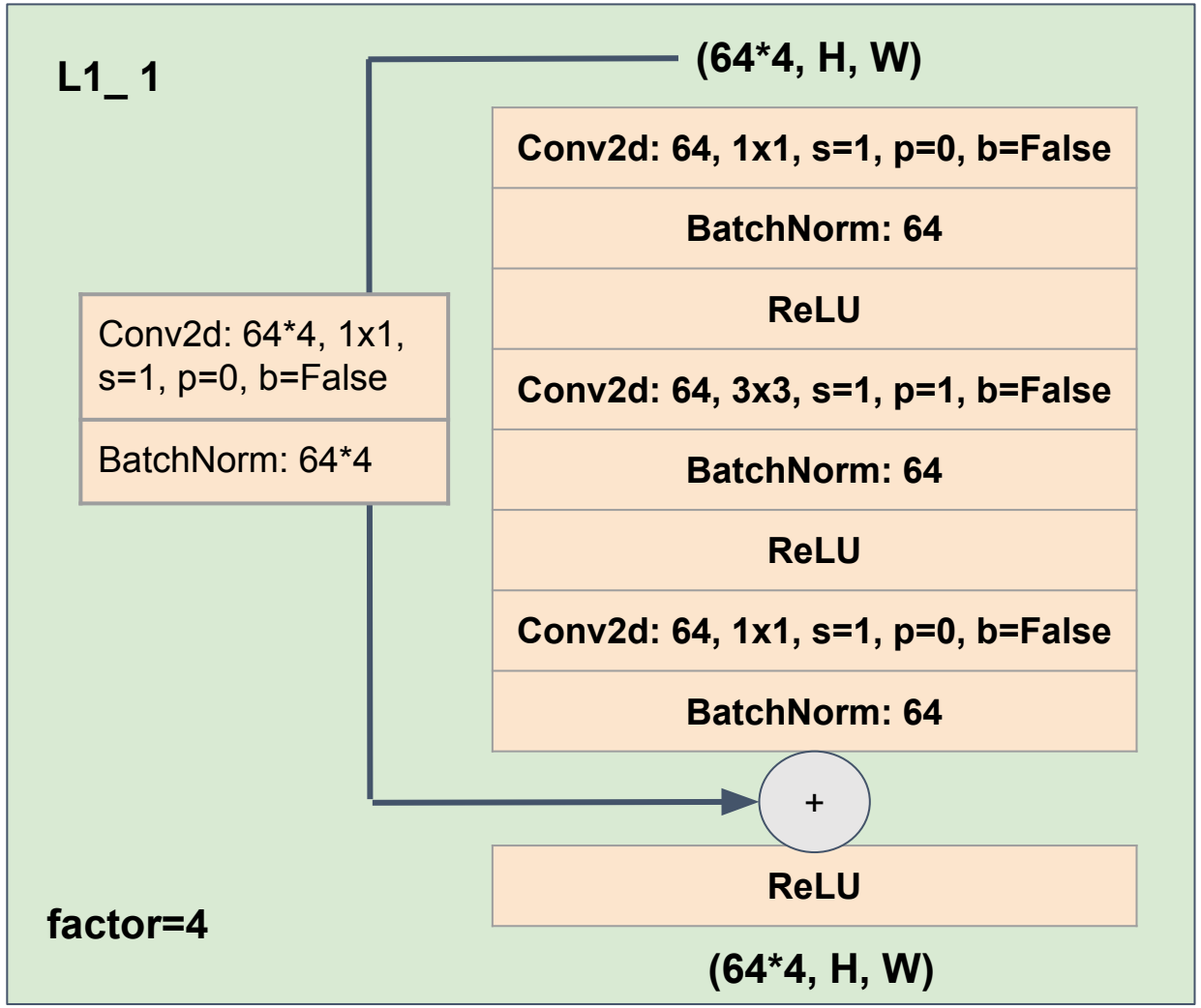
| Conv2d: 3, 64, 7x7, s=2, p=3, b=False |
|---|
| BatchNorm: 64 |
| MaxPool2d: 3x3, s=2, p=1 |
| Layer1: 64, H, W |
| Layer2: 128, H/2, W/2 |
| Layer3: 128, H/4, W/4 |
| Layer4: 512, H/8, W/8 |
| AdaptiveAvgPool2d |
| Linear: 512*factor, out=1000, b=True |

| BasicBlock | |
|---|---|
| resnet18 | resnet34 |
| 2 | 3 |
| 2 | 4 |
| 2 | 6 |
| 2 | 3 |

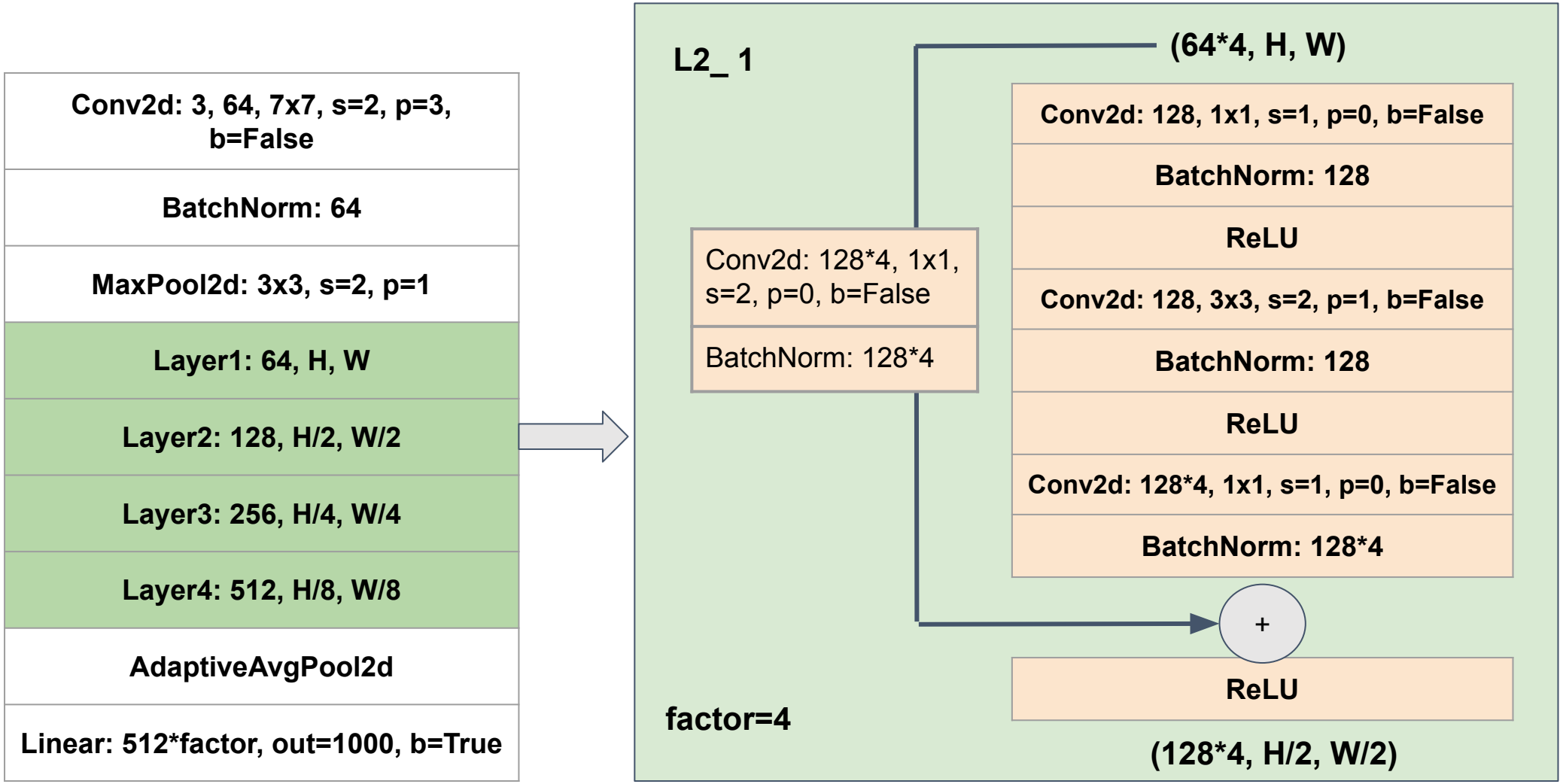
| BasicBlock | |
|---|---|
| resnet50 | resnet101 |
| 3 | 3 |
| 4 | 4 |
| 6 | 23 |
| 3 | 3 |

# Архитектура сети ResNet-50

| |
|---|
| Conv2d: 3, 64, 7x7, s=2, p=3, b=False |
| BatchNorm: 64 |
| MaxPool2d: 3x3, s=2, p=1 |
| Layer1: 64, H, W |
| Layer2: 128, H/2, W/2 |
| Layer3: 256, H/4, W/4 |
| Layer4: 512, H/8, W/8 |
| AdaptiveAvgPool2d |
| Linear: 512*factor, out=1000, b=True |

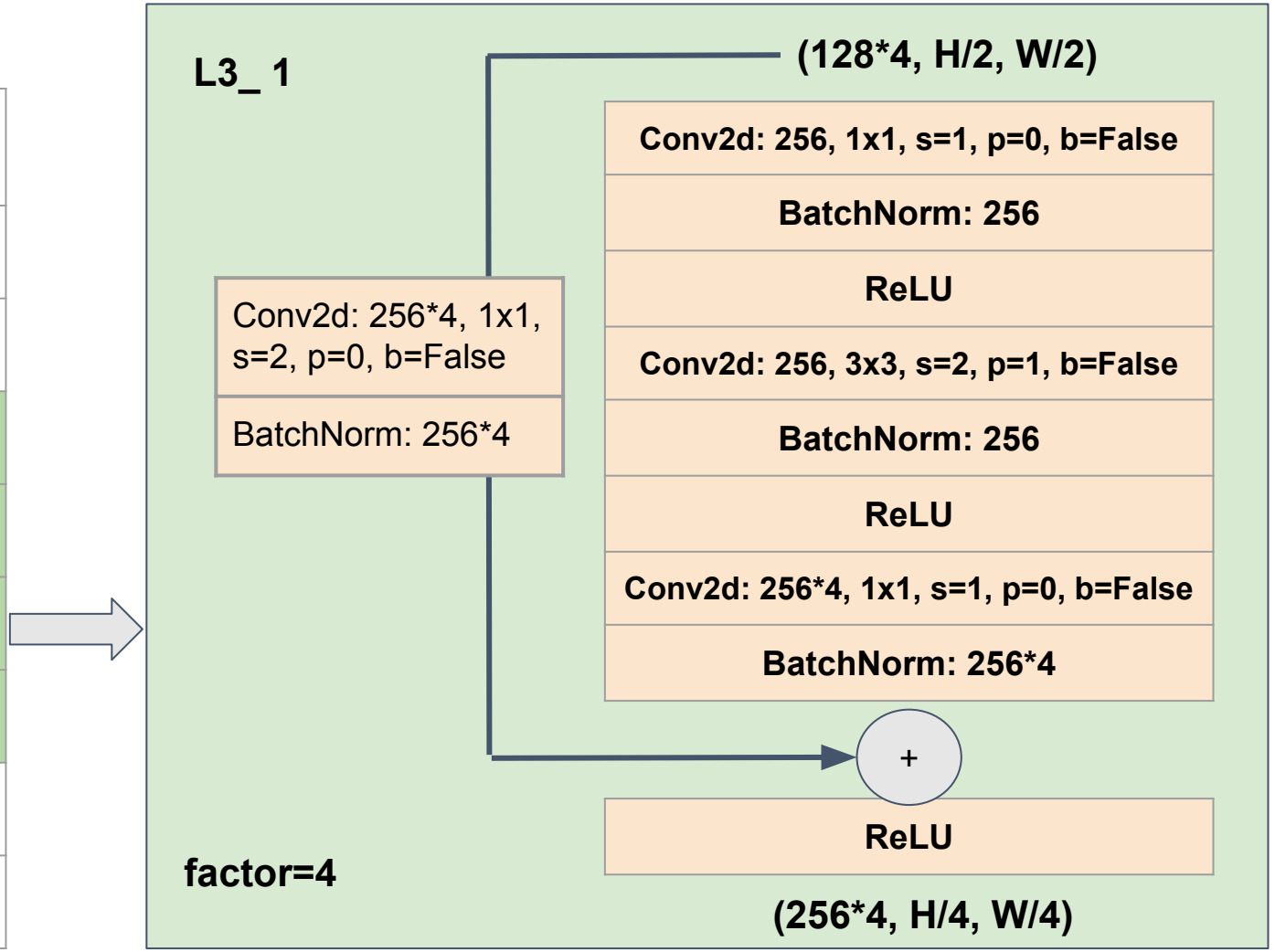


# Архитектура сети ResNet-50



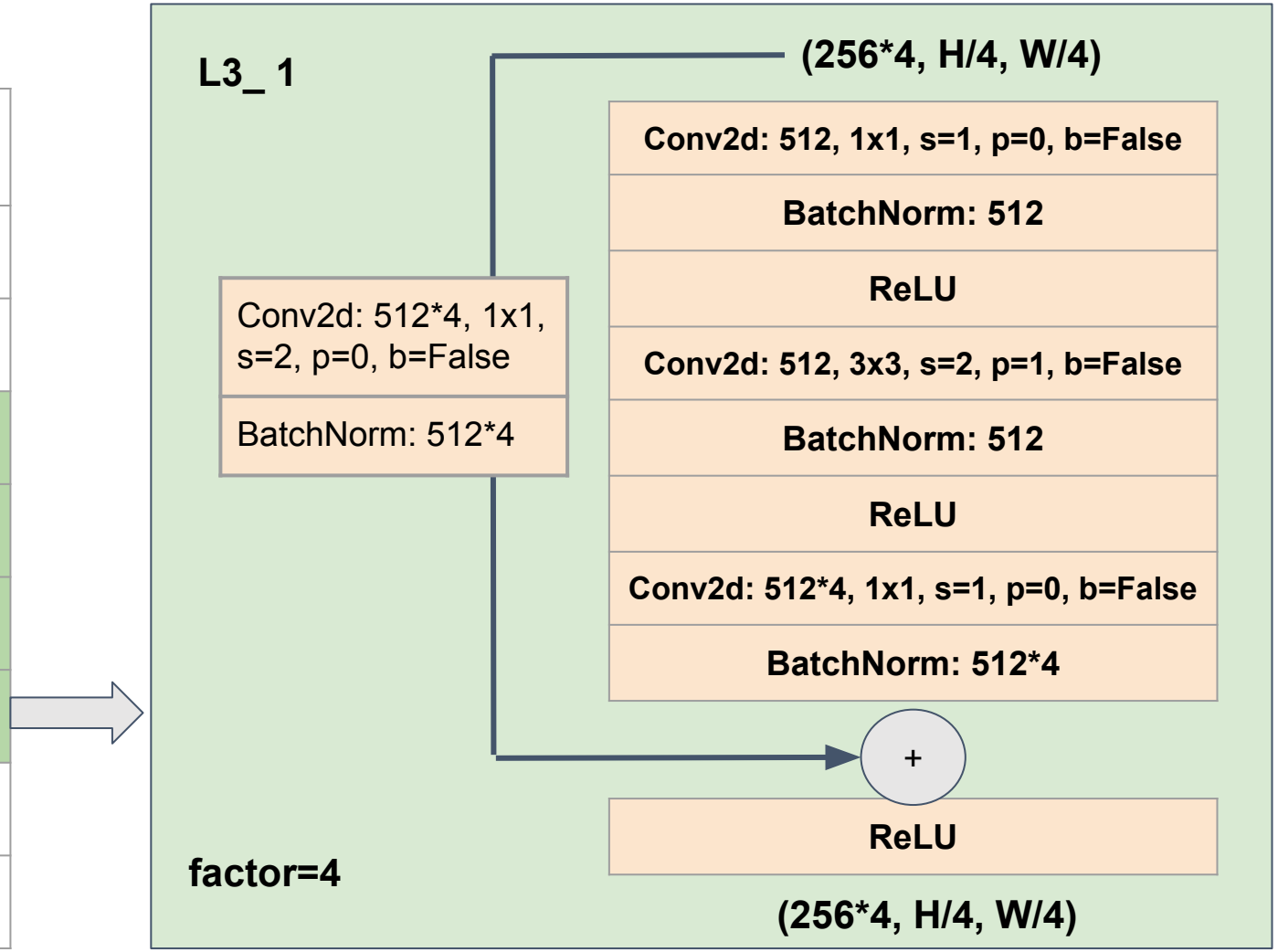
# Архитектура сети ResNet-50

| |
|---|
| Conv2d: 3, 64, 7x7, s=2, p=3, b=False |
| BatchNorm: 64 |
| MaxPool2d: 3x3, s=2, p=1 |
| Layer1: 64, H, W |
| Layer2: 128, H/2, W/2 |
| Layer3: 256, H/4, W/4 |
| Layer4: 512, H/8, W/8 |
| AdaptiveAvgPool2d |
| Linear: 512*factor, out=1000, b=True |



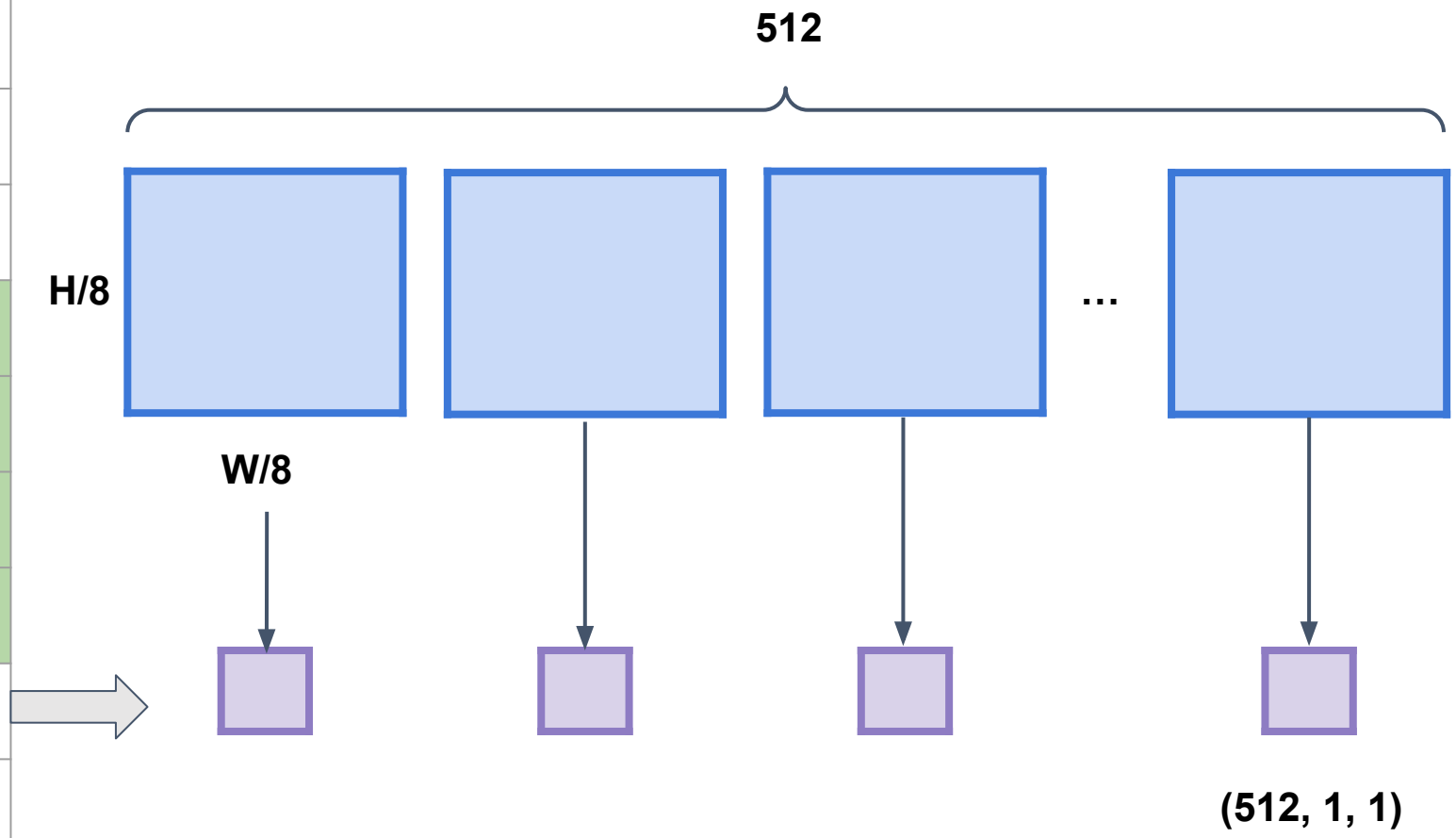
# Архитектура сети ResNet-50

| |
|---|
| Conv2d: 3, 64, 7x7, s=2, p=3, b=False |
| BatchNorm: 64 |
| MaxPool2d: 3x3, s=2, p=1 |
| Layer1: 64, H, W |
| Layer2: 128, H/2, W/2 |
| Layer3: 256, H/4, W/4 |
| Layer4: 512, H/8, W/8 |
| AdaptiveAvgPool2d |
| Linear: 512*factor, out=1000, b=True |



# Архитектура сети ResNet-18

| |
|---|
| Conv2d: 3, 64, 7x7, s=2, p=3, b=False |
| BatchNorm: 64 |
| MaxPool2d: 3x3, s=2, p=1 |
| Layer1: 64, H, W |
| Layer2: 128, H/2, W/2 |
| Layer3: 128, H/4, W/4 |
| Layer4: 512, H/8, W/8 |
| AdaptiveAvgPool2d |
| Linear: 512*factor, out=1000, b=True |



# Развертка сетей ResNet в ширину

